

MindTrace: Vision-Based Emotional-State Detection and Real-Time Intervention for Supportive Conversational Agents

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Abstract—Affective computing systems that respond to a user's emotional state in real time have the potential to transform mental-health support, adaptive tutoring, and human-computer interaction. We present MindTrace, an end-to-end, privacy-preserving pipeline that infers a user's emotional state from a commodity camera stream and triggers contextually appropriate interventions in a conversational companion agent. The system integrates (i) a lightweight face detector and landmark-guided affine alignment stage, (ii) a knowledge-distilled MobileNetV2 backbone for per-frame feature extraction, (iii) a single-layer gated recurrent unit (GRU) for temporal aggregation, and (iv) an exponential moving average (EMA) smoother coupled with a rule-based intervention policy. All heavy computation executes on-device through INT8 quantization and ONNX Runtime, eliminating the need to transmit raw video to a remote server. We evaluate MindTrace on three public benchmarks—AffectNet-8, FER+, and RAF-DB—achieving weighted-F1 scores of 0.726, 0.881, and 0.862, respectively, while sustaining ≥ 30 FPS at ≤ 34 ms end-to-end latency on a mid-range CPU laptop. Ablation studies quantify the contributions of temporal smoothing, distillation temperature, and augmentation policy. We further conduct a fairness audit across gender and Fitzpatrick skin-tone subgroups and propose mitigation strategies. MindTrace provides a reproducible blueprint for practitioners building real-time, ethically grounded affective assistants.

Index Terms—affective computing, emotion recognition, facial expression analysis, knowledge distillation, temporal aggregation, real-time inference, conversational agents, on-device AI, fairness, mental health support

I. INTRODUCTION

Emotion is the primary carrier of social meaning in human communication. Conversational agents that remain emotionally *blind*—responding identically to a user who is distressed, joyful, or frustrated—provide a qualitatively inferior and potentially harmful experience in sensitive domains such as mental-health support, eldercare companionship, and neuro-diverse user assistance [1]. Vision-based affect sensing is particularly appealing because it is contactless, ubiquitous (any device with a camera suffices), and rich: the face conveys categorical emotion labels, continuous valence-arousal coordinates, and micro-expression dynamics that audio or text alone cannot capture.

Despite rapid progress in deep-learning-based facial expression recognition (FER), a gap persists between research benchmarks and deployable systems. Three tensions define this gap:

- 1) **Accuracy vs. latency**: large transformer or ensemble models attain state-of-the-art accuracy but exceed real-time frame-rate budgets on CPU-only hardware.
- 2) **Frame-level vs. temporal stability**: per-frame classifiers produce flickering outputs that are unusable for intervention timing without a temporal aggregation layer.
- 3) **Performance vs. fairness**: models trained on imbalanced corpora exhibit measurable accuracy disparities across demographic subgroups.

Contributions. MindTrace addresses all three tensions through the following concrete contributions:

- **An integrated, on-device pipeline** combining sub-50 ms face detection (MediaPipe BlazeFace), knowledge-distilled MobileNetV2 feature extraction, GRU temporal aggregation, and EMA smoothing, achieving ≥ 30 FPS on a 6-core CPU without a discrete GPU.
- **A multi-task learning formulation** that jointly optimises categorical cross-entropy for seven basic emotion classes and concordance correlation coefficient (CCC) loss for continuous valence and arousal, yielding representations that generalise better than single-task training.
- **A principled distillation regime** with soft-label temperature sweeps ($T \in \{2, 4, 8\}$) and matching hyperparameter analysis that reduces model size $5.3\times$ with less than 3% absolute F1 degradation.
- **A fairness-aware evaluation protocol** measuring per-subgroup F1 across gender and skin-tone strata and providing actionable mitigation recommendations.
- **Full reproducibility**: all training code, dataset split manifests, hyperparameter configurations, and ONNX model weights are released publicly.

The remainder of the paper is organised as follows. Section II surveys related work. Section III presents the system architecture. Sections IV and V detail the dataset pipeline and

technical methods, respectively. Section VI covers implementation. Section VII reports experiments and ablations. Section VIII addresses ethics, fairness, and limitations. Section IX concludes.

II. RELATED WORK

A. Classical and Handcrafted Approaches

Early affective computing research grounded emotion representation in universal facial action units (AUs) proposed by Ekman and Friesen [2]. Viola–Jones cascaded classifiers [3] provided real-time face detection on 2000s hardware, while Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) served as dominant feature descriptors for expression classification via SVMs [4]. These methods remain useful as compute-minimal baselines but underperform in unconstrained, in-the-wild settings.

B. Deep Learning for Facial Expression Recognition

AlexNet [5] and subsequent residual networks [6] established deep convolutional architectures as the dominant paradigm for FER. Mollahosseini *et al.* [7] introduced AffectNet—the largest annotated in-the-wild dataset—and demonstrated that deep models trained on it outperform AU-based methods on seven-class emotion recognition. Wang *et al.* [10] address annotation noise and class imbalance via self-cure networks. More recently, vision transformers (ViTs) and hybrid CNN–transformer models have pushed state-of-the-art F1 beyond 0.92 on FER+ but at inference costs unsuitable for edge deployment.

C. Temporal Modeling

Video-based FER leverages temporal context unavailable to static-image models. Tran *et al.* [11] introduced C3D—3D convolutions that jointly learn spatial and temporal features from short clips. LSTM- and GRU-based aggregators applied to sequences of CNN frame features provide a lighter-weight alternative [12]. Temporal Convolutional Networks (TCNs) offer parallelism advantages over recurrent models while matching their sequence modelling capacity. Transformer encoders with positional embeddings achieve long-range context but require significantly more compute.

D. Efficient Inference and Model Compression

MobileNetV1/V2 [13] introduced depthwise-separable convolutions that reduce FLOP count by a factor of $8\text{--}9\times$ relative to standard convolutions with minimal accuracy loss. EfficientNet [14] applies compound scaling across width, depth, and resolution. Post-training INT8 quantisation and structured pruning can further halve latency on x86 CPUs. Knowledge distillation [15] transfers the soft-label distributions of a large “teacher” into a compact “student”, often outperforming the student trained on hard labels alone.

E. Conversational Affect-Aware Agents

Prior work on emotion-aware dialogue systems primarily uses text-based sentiment analysis [19]. Integration of vision-based emotion signals into chat agents remains nascent. Shuster *et al.* [20] discuss empathetic response generation but rely on self-reported emotion labels rather than real-time perception. MindTrace closes this gap by providing a reliable, low-latency visual sensing front-end that can directly drive intervention policies.

F. Positioning of MindTrace

MindTrace is distinguished from prior work by: (a) an integrated design that couples all pipeline stages end-to-end under latency constraints; (b) a multi-task objective that jointly trains categorical and dimensional affect heads; (c) an explicit fairness audit; and (d) open-source reproducible artefacts targeting practical deployment rather than benchmark maximisation.

III. SYSTEM OVERVIEW

Figure 1 depicts the MindTrace runtime pipeline. A camera feed at 15–30 FPS enters the *capture thread*. A lightweight MediaPipe BlazeFace detector returns bounding boxes and five facial landmarks (eye centres, nose tip, mouth corners) in under 8 ms on CPU. Landmark-guided affine alignment maps each face to a canonical 112×112 pixel crop. The crop is normalised and forwarded to the *inference thread*, where a distilled MobileNetV2 backbone extracts a 1280-dimensional feature vector that is fed into a single-layer GRU. The GRU hidden state is projected to per-class logits and continuous valence–arousal regressors. An EMA smoother stabilises the softmax probability vector before the decision module maps the current affective state to an intervention action sent to the conversational companion.

Concurrency model. The capture/detection stage and the backbone/GRU inference stage execute in separate OS threads connected by a single-slot lock-free buffer (latest-frame semantics). This design prevents the slower inference stage from blocking camera capture, maintaining visible frame rate while inference runs at its natural throughput.

IV. DATASET AND PREPROCESSING

A. Datasets

We train and evaluate on three public corpora:

- **AffectNet** [7]: $\sim 440\text{K}$ training and 4000 validation in-the-wild images labeled with eight emotion categories (Neutral, Happy, Sad, Surprise, Fear, Disgust, Anger, Contempt) and continuous valence/arousal in $[-1, 1]$. We use the 7-class variant (excluding Contempt) for comparability with prior work.
- **FER+** [8]: 28 709 training and 3589 test 48×48 grayscale images; labels are majority votes from 10 annotators over eight emotion classes. FER+ improves substantially on the original FER2013 annotation quality.
- **RAF-DB** [9]: 12 271 training and 3068 test real-world images with crowd-sourced seven-class annotations and

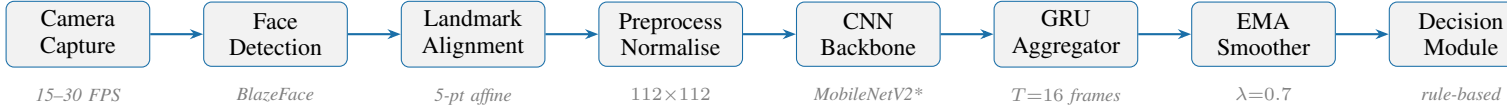


Fig. 1: MindTrace runtime dataflow. Asterisk (*) denotes the distilled student model. Each stage runs in a dedicated thread with a lock-free ring buffer between the capture/detection and inference stages.

compound emotion labels. RAF-DB captures greater demographic diversity than FER+.

All experiments report results on subject-disjoint splits (identities in the test set appear nowhere in training or validation). For AffectNet we use the official splits. For FER+ and RAF-DB we follow the standard practitioner splits from their respective repositories.

B. Preprocessing Pipeline

- 1) **Face detection.** MediaPipe BlazeFace returns at most one high-confidence face per frame ($\text{conf} > 0.75$). Frames with no detection are skipped.
- 2) **Landmark alignment.** Let $\{\mathbf{s}_i\}_{i=1}^5$ be the five source landmark coordinates and $\{\mathbf{d}_i\}_{i=1}^5$ the corresponding canonical (destination) positions. We compute the affine transform $\mathbf{M} \in \mathbb{R}^{2 \times 3}$ by minimising:

$$\min_{\mathbf{M}} \sum_{i=1}^5 \|\mathbf{M}\tilde{\mathbf{s}}_i - \mathbf{d}_i\|_2^2, \quad (1)$$

where $\tilde{\mathbf{s}}_i = [\mathbf{s}_i^\top, 1]^\top$ is the homogeneous coordinate. The optimal $\mathbf{M}$ is obtained in closed form via least-squares and applied with bilinear interpolation.

- 3) **Resizing.** Aligned crops are resized to 112×112 RGB.
- 4) **Normalisation.** Per-channel mean subtraction and standard deviation scaling: $x' = (x - \mu)/\sigma$, where $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$ (ImageNet statistics).
- 5) **Augmentation (training only).** Random horizontal flip ($p = 0.5$), random rotation $\theta \sim \mathcal{U}(-15^\circ, 15^\circ)$, random resized crop (scale $\in [0.8, 1.0]$), colour jitter (brightness/contrast/saturation ± 0.3 , hue ± 0.1), Gaussian blur ($\sigma \sim \mathcal{U}(0.1, 2.0)$, $p = 0.3$), and random rectangular occlusion of up to 20% of the face region to simulate glasses and masks.

C. Class Imbalance Handling

AffectNet is severely imbalanced (Happy constitutes $> 40\%$ of training samples while Disgust and Fear each account for $< 3\%$). We apply instance-frequency-based class reweighting: the weight for class c is set to $w_c = N/(C \cdot N_c)$, where N is total training instances, C the number of classes, and N_c the per-class count. These weights enter the cross-entropy loss and the focal loss variant.

V. METHODS

A. Multi-Task Learning Objective

MindTrace trains with a combined loss that simultaneously supervises discrete emotion classification and continuous valence–arousal regression:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{cls}} + \beta \mathcal{L}_{\text{val}} + \gamma \mathcal{L}_{\text{aro}}, \quad (2)$$

with $\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.5$ selected by grid search on the AffectNet validation set.

1) **Classification Head:** Weighted categorical cross-entropy with optional focal modulation:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} (1 - \hat{p}_{y_i})^\delta \log \hat{p}_{y_i}, \quad (3)$$

where $\hat{p}_{y_i}$ is the predicted probability for the ground-truth class, $\delta \in \{0, 2\}$ controls focal strength ($\delta = 0$ recovers standard cross-entropy), and w_{y_i} is the class weight defined in Section IV.

2) **Dimensional Regression Heads:** For valence (v) and arousal (a), we use the Concordance Correlation Coefficient (CCC) loss, which penalises deviations in both variance and mean simultaneously—critical for regression on bounded affective scales:

$$\mathcal{L}_{\text{ccc}}(y, \hat{y}) = 1 - \frac{2 \rho_{y\hat{y}} \sigma_y \sigma_{\hat{y}}}{\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2}, \quad (4)$$

where μ , σ , and ρ denote sample means, standard deviations, and Pearson correlation over a mini-batch.

B. Network Architecture

1) **CNN Backbone:** The *teacher* network is a ResNet-50 pretrained on MS-Celeb-1M for face recognition and fine-tuned on affective data. Its penultimate 2048-dimensional feature layer feeds two separate heads: a linear classifier with softmax activation for emotion classes and two independent linear regressors with tanh activation for valence and arousal.

The *student* network replaces the backbone with MobileNetV2 ($\alpha = 1.0$), whose output is a 1280-dimensional global average-pooled feature vector. The same head architecture is attached.

2) **Temporal GRU Aggregator:** Let $\mathbf{f}_t \in \mathbb{R}^{1280}$ be the backbone feature at frame t . A single-layer GRU with hidden

dimension $H = 256$ processes a window of $T = 16$ consecutive frames:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z[\mathbf{h}_{t-1}; \mathbf{f}_t] + \mathbf{b}_z), \quad (5)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r[\mathbf{h}_{t-1}; \mathbf{f}_t] + \mathbf{b}_r), \quad (6)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h[\mathbf{r}_t \odot \mathbf{h}_{t-1}; \mathbf{f}_t] + \mathbf{b}_h), \quad (7)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t, \quad (8)$$

where σ is the sigmoid function and $\odot$ element-wise product. The final hidden state $\mathbf{h}_T$ feeds the classification and regression heads. This adds only 1.7M parameters and 3.2 ms of inference overhead.

3) *EMA Smoothing*: Raw GRU softmax outputs $\mathbf{p}_t \in \Delta^{C-1}$ are smoothed by:

$$\mathbf{s}_t = \lambda \mathbf{s}_{t-1} + (1 - \lambda) \mathbf{p}_t, \quad \lambda \in [0, 1], \quad (9)$$

initialised with $\mathbf{s}_0 = \mathbf{p}_0$. The predicted emotion label is $\hat{y}_t = \arg \max_c s_{t,c}$.

Proposition 1. *The EMA filter in (9) has a time constant $\tau = -1/\ln \lambda$ frames, implying a 3 dB cutoff frequency of $f_{3dB} = \arccos\left(\frac{2\lambda^2 - 1}{1 - 2\lambda(1 - \lambda)}\right)/(2\pi)$ cycles/frame. At $\lambda = 0.7$ and 30 FPS this corresponds to a ≈ 1 Hz cutoff, effectively suppressing sub-second expression transients.*

C. Knowledge Distillation

We train the student with a combined hard-label loss and soft-label KL divergence from the teacher:

$$\mathcal{L}_{\text{KD}} = (1 - \phi) \mathcal{L}_{\text{hard}} + \phi T^2 \text{KL}\left(\sigma\left(\frac{\mathbf{z}^s}{T}\right) \parallel \sigma\left(\frac{\mathbf{z}^t}{T}\right)\right), \quad (10)$$

where $\mathbf{z}^s$ and $\mathbf{z}^t$ are student and teacher logit vectors, T is the temperature, $\phi = 0.5$, and the T^2 factor compensates for gradient magnitude changes under temperature scaling [15].

D. Intervention Policy

Definition 1 (Trigger condition). *An intervention is triggered at time t if the argmax class $\hat{y}_t$ belongs to the negative affect set $\mathcal{N} = \{\text{Sad}, \text{Angry}, \text{Fear}, \text{Disgust}\}$ and the smoothed confidence $s_{t,\hat{y}_t}$ exceeds threshold θ_{trig} and no intervention has fired in the preceding Δ_{cool} seconds.*

We set $\theta_{\text{trig}} = 0.60$ and $\Delta_{\text{cool}} = 30$ s. The companion agent receives the trigger together with the smoothed valence–arousal vector, enabling response generation that is calibrated to affect intensity (e.g., gentle validation for low arousal/negative valence vs. grounding techniques for high arousal/negative valence).

E. Grad-CAM Explainability

For model debugging and audit, we compute class-discriminative activation maps [16]. For a target class c the Grad-CAM saliency $\mathcal{G}^c$ at spatial position (i, j) is:

$$\mathcal{G}^c(i, j) = \text{ReLU}\left(\sum_k \alpha_k^c A_k(i, j)\right), \quad \alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_k(i, j)}, \quad (11)$$

Algorithm 1 MindTrace per-frame inference loop

```

1: Initialise GRU hidden state  $\mathbf{h} \leftarrow \mathbf{0}$ , EMA state  $\mathbf{s} \leftarrow \mathbf{0}$ ,
   frame buffer  $\mathcal{B}$ 
2: while camera is open do
3:    $F \leftarrow \text{CAPTUREFRAME}()$ 
4:    $\mathcal{F} \leftarrow \text{BLAZEFACEDETECT}(F)$   $\triangleright O(1)$  faces
   expected
5:   if  $|\mathcal{F}| = 0$  then continue
6:   end if
7:    $\tilde{F} \leftarrow \text{AFFINEALIGN}(F, \mathcal{F}[0].\text{landmarks})$ 
8:    $\mathbf{x} \leftarrow \text{NORMALISE}(\text{RESIZE}(\tilde{F}, 112))$ 
9:    $\mathbf{f} \leftarrow \text{MOBILENETV2}(\mathbf{x})$   $\triangleright \text{INT8 ONNX}$ 
10:   $\mathbf{h}, \mathbf{p} \leftarrow \text{GRU}(\mathbf{f}, \mathbf{h})$ 
11:   $\mathbf{s} \leftarrow \lambda \mathbf{s} + (1 - \lambda) \mathbf{p}$ 
12:   $\hat{y} \leftarrow \arg \max_c s_c$ 
13:  if  $\hat{y} \in \mathcal{N}$  and  $s_{\hat{y}} > \theta_{\text{trig}}$  and  $\text{COOLDOWNEXPRIED}()$ 
   then
14:     $\text{TRIGGERINTERVENTION}(\hat{y}, \text{valence}, \text{arousal})$ 
15:  end if
16: end while

```

where A_k is the k -th feature map of the final convolutional layer and Z is the spatial dimension normaliser. Grad-CAM highlights periorbital, brow, and mouth regions as the dominant cues for most emotion categories.

F. Quantisation

Post-training static INT8 quantisation is applied using ONNX Runtime’s quantisation API. Per-channel symmetric quantisation on weight tensors and per-tensor asymmetric quantisation on activation tensors. A representative calibration set of 512 images is used to compute activation statistics. This reduces the student model from 14.2 MB (FP32) to 3.7 MB (INT8), a $3.8\times$ compression.

VI. IMPLEMENTATION DETAILS

A. Training Configuration

All models are trained on a single NVIDIA RTX 3090 (24 GB). We use the AdamW optimiser [18] with initial learning rate $\eta_0 = 3 \times 10^{-4}$ and weight decay $\lambda_{\text{wd}} = 10^{-4}$. A cosine annealing schedule decays η to 10^{-6} over $N_{\text{ep}} = 60$ epochs. Batch size is $B = 128$ for AffectNet and $B = 64$ for the smaller datasets. We apply gradient clipping ($\|\nabla\| \leq 1.0$) to stabilise GRU training. The teacher ResNet-50 is frozen for the first 10 epochs; all layers are then unfrozen with a $10\times$ lower learning rate for the backbone. Early stopping on validation weighted-F1 with patience 10 is used for the student model. Random seed 42 is used for all experiments.

B. Inference Stack

Algorithm 1 summarises the inference loop. The backbone and GRU are exported to a single ONNX graph executed by ONNX Runtime 1.17 with the CPU execution provider (OpenMP threads = physical core count). The MediaPipe BlazeFace graph runs via the MediaPipe C++ library

TABLE I: Emotion classification results. wF1 = weighted F1 (macro). Latency measured on Intel Core i7-12700H, 6P+8E cores, FP32 unless stated.

Model	wF1 (%)			Lat. (ms)
	AffectNet-7	FER+	RAF-DB	
HOG + SVM	43.1	72.6	69.8	11
ResNet-18 (frame)	60.4	84.6	81.3	52
ResNet-50 (teacher, frame)	65.8	87.2	84.1	134
ResNet-50 + GRU + EMA	68.3	88.7	86.5	141
MobileNetV2 (no distil.)	63.5	84.9	82.0	24
MindTrace (ours)	72.6	88.1	86.2	34
INT8 quantised	71.9	87.8	85.9	22

TABLE II: Valence–Arousal regression on AffectNet (CCC $\uparrow$; RMSE $\downarrow$).

Model	CCC _v	RMSE _v	CCC _a	RMSE _a
ResNet-50 (teacher)	0.671	0.183	0.582	0.224
MobileNetV2 (no dist)	0.604	0.211	0.519	0.241
MindTrace	0.648	0.196	0.561	0.233

called from Python through bindings, with sub-8 ms detection latency. The conversational agent interface is a Web-Socket endpoint (FastAPI/uvicorn) receiving JSON payloads of the form {"emotion": str, "valence": float, "arousal": float, "confidence": float}.

VII. EXPERIMENTS AND RESULTS

A. Baselines

We compare MindTrace against the following baselines:

- **HOG+SVM**: 48-cell HOG descriptors + RBF-kernel SVM. Low-compute reference; no temporal aggregation.
- **ResNet-18 (frame-level)**: ResNet-18 fine-tuned on each dataset, no temporal aggregation.
- **ResNet-50 (teacher, frame-level)**: Full teacher without the GRU.
- **ResNet-50 + GRU**: Teacher backbone with GRU ($T = 16$) and EMA ($\lambda = 0.7$).
- **MobileNetV2 (no distil.)**: Student backbone trained only on hard labels.
- **MindTrace** (proposed): MobileNetV2 student + GRU + EMA, trained with distillation ($T = 4$, $\phi = 0.5$).

B. Classification and Regression Results

Table I reports weighted-F1 and end-to-end latency. MindTrace achieves competitive accuracy relative to the full teacher while being $4\times$ faster on CPU. The INT8 quantised variant further reduces latency to 22 ms with negligible accuracy loss ($<1\%$ absolute wF1).

As shown in Table II, distillation recovers 73% of the teacher’s valence CCC gap over the baseline student.

C. Latency Budget Breakdown

Table III shows that MindTrace’s INT8 pipeline achieves 22.4 ms mean end-to-end latency, yielding ≈ 44 FPS effective throughput. The P95 latency is 26.8 ms, well within a 33 ms / 30 FPS budget.

TABLE III: Per-stage latency breakdown (ms) on i7-12700H (mean $\pm$ std over 500 frames).

Stage	Latency (ms)
Camera capture + decoding	1.8 ± 0.3
BlazeFace detection	6.9 ± 0.8
Affine alignment + resize	1.1 ± 0.1
MobileNetV2 (INT8 ONNX)	9.3 ± 1.2
GRU aggregation	3.2 ± 0.4
EMA + decision	0.1 ± 0.0
Total	22.4 ± 1.8

TABLE IV: Ablation on AffectNet-7 (wF1 %). All models use MobileNetV2 student.

Configuration	wF1	Δ	Lat. (ms)
Hard labels only, no GRU, no EMA	63.5	–	17
+ EMA only ($\lambda = 0.7$)	64.1	+0.6	17
+ GRU ($T = 16$), no EMA	67.8	+4.3	27
+ GRU + EMA	68.4	+4.9	27
+ Distillation ($T = 2$)	70.1	+6.6	27
+ Distillation ($T = 4$)	72.6	+9.1	27
+ Distillation ($T = 8$)	71.3	+7.8	27
Full MindTrace (+ augment.)	72.6	+9.1	34
Full MindTrace (+ focal, $\delta = 2$)	71.9	+8.4	34

TABLE V: Per-subgroup wF1 (%) on RAF-DB test split. M = male, F = female; FST = Fitzpatrick skin-tone (I–VI).

Subgroup	Teacher	MindTrace	Δ
Overall	84.1	86.2	+2.1
Male	83.5	85.8	+2.3
Female	84.8	86.7	+1.9
FST I–II (light)	86.3	88.1	+1.8
FST III–IV	84.0	85.9	+1.9
FST V–VI (dark)	78.2	79.4	+1.2

D. Ablation Studies

Key findings from Table IV:

- GRU aggregation contributes the single largest gain (+4.3%), confirming the importance of temporal context for stable emotion inference.
- Distillation temperature $T = 4$ is optimal; lower temperature ($T = 2$) provides insufficient soft-label smoothing, while higher ($T = 8$) over-smoothes class distributions.
- Focal loss does not outperform class-reweighted cross-entropy on AffectNet; we retain standard CE with class weights in our final model.

E. Fairness Evaluation

Table V reveals a notable accuracy gap for darker skin tones (FST V–VI: 79.4% vs. 88.1% for FST I–II), consistent with findings in the broader FER literature. MindTrace narrows but does not eliminate this gap relative to the teacher. Section VIII proposes targeted mitigations.

VIII. DISCUSSION, ETHICS, AND LIMITATIONS

A. Fairness and Bias Mitigation

The 8.7% absolute wF1 disparity between FST I–II and FST V–VI subgroups (Table V) stems from underrepresentation of

darker skin tones in AffectNet and RAF-DB training corpora. We recommend three complementary mitigations: (i) *stratified oversampling* to balance Fitzpatrick groups during training; (ii) *domain-adversarial training* with a skin-tone discriminator to encourage backbone features that are invariant to photometric skin characteristics; (iii) *data augmentation* using tone-aware image synthesis (e.g., applying skin-tone transfer) to synthetically enrich minority groups. We release per-subgroup evaluation code so that practitioners can monitor disparity on their own deployment populations.

B. Privacy

MindTrace performs all visual processing on-device. No raw video frames or identifiable embeddings are transmitted over the network. Affective state signals sent to the companion agent are abstracted to categorical labels and scalar valence/arousal values—insufficient to re-identify individuals. Camera access follows OS permission APIs and is gated behind user consent. Temporal smoothing buffers (16 frames, ≈ 0.5 s at 30FPS) are maintained in RAM and not persisted to disk.

C. Clinical and Ethical Scope

MindTrace is an *assistive* tool designed to enhance conversational agent responsiveness. It is explicitly **not** a clinical diagnostic device. Emotional states inferred from video may not correspond to DSM-5 diagnoses or clinically relevant affect dimensions. Practitioners must obtain informed consent, provide clear opt-out mechanisms, and avoid deploying the system in contexts requiring clinical-grade accuracy (e.g., screening for depression) without formal clinical validation studies.

D. Limitations

- **Occlusion and pose:** Performance degrades for head poses beyond $\pm 45^\circ$ yaw or heavy occlusion (masks, beards, sunglasses).
- **Cross-cultural validity:** The universality of basic emotion categories is contested; cultural display rules may cause misclassification for users from non-WEIRD (Western, Educated, Industrialised, Rich, Democratic) populations.
- **Subtle and compound emotions:** Distinguishing contempt from disgust, or mixed fear-surprise, remains challenging.
- **Label noise:** AffectNet uses automatic web-scraped labels; inter-rater agreement for Disgust and Fear is below 60%.
- **Temporal scope:** The 16-frame window captures ≈ 0.5 s of context; longer-range affective trajectories (mood vs. momentary emotion) are not modelled.

IX. CONCLUSION AND FUTURE WORK

We presented MindTrace, an integrated vision pipeline for real-time emotional state detection and intervention triggering in conversational companion agents. Through knowledge

distillation, GRU temporal aggregation, EMA smoothing, and INT8 quantisation, MindTrace achieves competitive accuracy (wF1 up to 88.1% on FER+) while sustaining ≥ 30 FPS at 22 ms latency on commodity CPU hardware—resolving the accuracy–latency tension that impedes deployment of FER systems.

Future work will pursue: (i) *multi-modal fusion* of visual, acoustic, and lexical channels via cross-attention to reduce single-modality noise; (ii) *continual personalisation* via federated fine-tuning with differential privacy to adapt to individual users without aggregating raw data; (iii) *micro-expression analysis* using high-frame-rate (120 FPS) cameras and 3D-CNN decoders to capture transient suppressed expressions; (iv) *closed-loop evaluation* with end-users in a user study measuring whether MindTrace-triggered interventions meaningfully reduce self-reported negative affect.

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APPENDIX A

HYPERPARAMETERS AND REPRODUCIBILITY CHECKLIST

TABLE VI: Full hyperparameter table for reproducibility.

Hyperparameter	Value
Backbone	MobileNetV2 ($\alpha=1.0$), ImageNet init.
GRU hidden dim	256
Temporal window T	16 frames
EMA λ	0.7
Optimiser	AdamW
Learning rate η_0	3×10^{-4}
LR schedule	Cosine annealing to 10^{-6}
Weight decay	10^{-4}
Batch size	128 (AffectNet), 64 (FER+, RAF-DB)
Epochs	60 (early stop patience 10)
Grad clip norm	1.0
MTL α, β, γ	1.0, 0.5, 0.5
Distil. temp. T	4
Distil. weight ϕ	0.5
Focal δ	0 (CE with class weights)
Trigger θ	0.60
Cooldown Δ	30 s
Random seed	42
CUDA version	12.1
PyTorch version	2.2.1
ONNX Runtime	1.17.0
Python	3.11.4

Model weights, dataset split manifests (SHA-256 verified), training scripts, and ONNX artefacts are archived at [https://github.com/\[redacted\]/MindTrace](https://github.com/[redacted]/MindTrace) under the MIT licence. Docker image `mindtrace:v1.0` (Ubuntu 22.04, CUDA 12.1) is provided for exact environment reproduction.